Project 2

"ProxiSpeak": *Geospatial Proximity Audio Ecosystem*

Overview:

ProxiSpeak splits the virtual office into zones, and the zone type decides who can hear whom. In open areas, distance alone sets the volume. In meeting rooms, focus rooms and stages, zone rules override distance. The design borrows ideas from Gather, Kumospace and WorkAdventure, and every rule is stored as data so the office layout can change without code changes.

Zone Type:

The office map has five zone types, each tied to a place on the floor plan.

Zone	Where on the map	Purpose
Open proximity	Open desks, lounge	Walk up and chat; volume depends on distance
Private	Meeting room, phone booth	Conversation stays inside the zone
Silent	Focus room	Nobody talks, even when standing close
Stage	Stage	One speaker is heard by the whole room
Lobby	Entrance	Spawn point for new users; behaves like open proximity

Audio rules per zone:

Each zone type answers one question: can person A hear person B?

Zone	A can hear B when	Volume
Open proximity	B is within the earshot radius (100 px) and not behind a wall	Falls off with distance
Private	A and B are in the same private zone	Full volume inside, silent outside
Silent	Never	Muted for everyone in the zone
Stage	B is on stage and A is in the same room	Full volume, broadcast
Lobby	Same as open proximity	Falls off with distance

People outside a private zone cannot hear in, and people inside cannot hear out, even if they stand against the wall.

How the server decides who can hear whom

Every avatar move runs through the same four steps on the Node.js server.

1. *The client sends its new X/Y position over Socket.io.*
2. *The in-memory spatial grid returns avatars within earshot, checking only the neighbouring 100 px cells.*
3. *The zone check looks up the zone each avatar stands in and applies the rules from the table above.*
4. *The server compares the result with the current peer list. It tells both browsers to connect when a pair becomes audible and to disconnect when they separate.*

   To stop connections flickering at the edge of earshot, connect at 100 px but disconnect only at 140 px.

Scaling notes

Zones keep each person's connections small, but big rooms need a different network design than small chats.

- *Small groups (about 4 people or fewer): direct peer-to-peer WebRTC audio, so the server never touches audio.*
- *Stage and large meeting rooms: a media server (an SFU such as LiveKit or mediasoup), because a full mesh does not scale to a 50-person all-hands. WorkAdventure makes the same switch when a bubble grows past 4 people.*
- *Positions live in server memory, not in MongoDB, so proximity checks stay well under 20 ms. MongoDB holds zones, obstacles and periodic snapshots.*
- *Several Node servers share state through Redis pub/sub, with one room kept on one server.*

